

Derbyshire Shared Care Pathology Guidelines

Heart Failure

Purpose of Guideline

This guideline covers the investigation and referral of patients with suspected heart failure and refers to the NICE guidelines NG106 (Chronic Heart Failure) and CG187 (Acute Heart Failure).

Natriuretic Peptides

The assay used in Derby, Chesterfield and Burton laboratories measures NT-proBNP (N-terminal pro-Brain Natriuretic Peptide) and all levels in this guideline are for NT-proBNP.

Measuring NT-proBNP greatly improves the ability to rule out heart failure before referring for ECHO,

Diagnosing Heart Failure

Acute Heart Failure:

NICE CG187 states that in patients presenting with new suspected acute heart failure, NT-proBNP <300 ng/L rules out the diagnosis of acute heart failure.

Chronic Heart Failure:

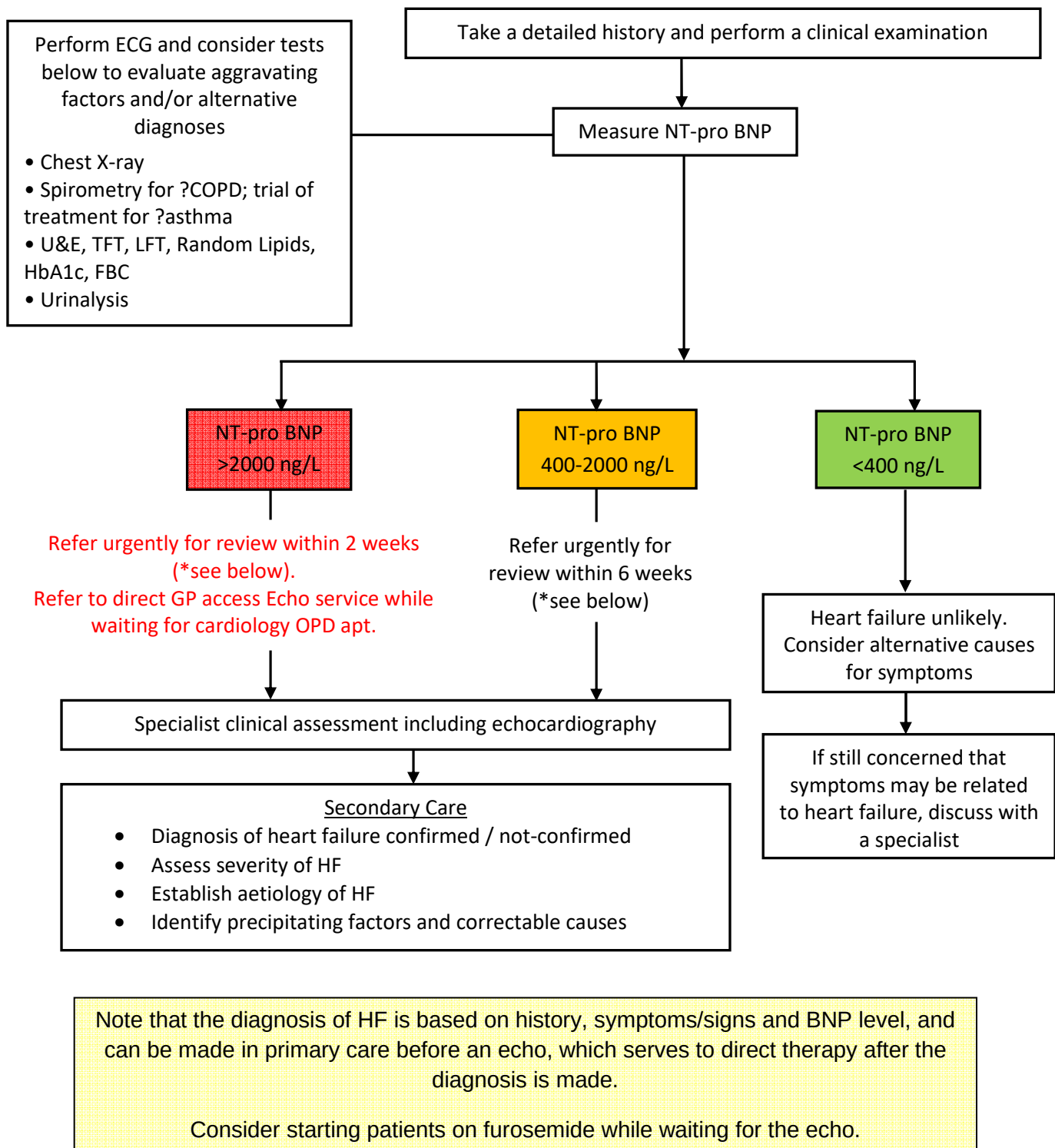
The following cut-offs are taken from NICE NG106, however it is **ESSENTIAL** to consider clinical context when interpreting results and prior to any referral. See Considerations for referral on page 3.

NT-proBNP (ng/L)	Interpretation
<400	In the absence of heart failure therapy, Heart Failure is an unlikely cause for the presentation
400 - 2000	Moderately raised, refer for review within 6 weeks (see "Considerations for referral" on page 3)
>2000	Significantly raised, refer for review within 2 weeks (see "Considerations for referral" on page 3)

The applicability of this interpretation when the patient has renal dysfunction is currently uncertain - suggest further advice is sought.

Treatment of heart failure can bring NT-proBNP into the normal range. Therefore, when questioning the diagnosis in patients on treatment, either stop their medication for three days before taking the test or leave on treatment and get an ECHO.

See the following flow chart for diagnosis and referral of patients with suspected chronic heart failure:



***Considerations for referral**

Whilst the flowchart above is aligned with NICE heart failure guidelines, the demand for echo services means that the 2 and 6 week review targets are rarely achievable. Therefore, clinicians should use clinical judgement in considering whether urgent echo is actually required. For example, it is very common for elderly patients with atrial fibrillation presenting with non-limiting breathlessness to have a NT-proBNP of >2000 ng/L. Echocardiography often shows normal LV systolic function and therefore has little impact on management. On the other hand, younger patients with severe valve disease or severe LV systolic dysfunction may have a NT-proBNP <2000 ng/L. Early echocardiography could be life-saving. If you have any doubt as to next steps for your patient, please use Consultant Connect or Advice and Guidance to discuss your patient with a consultant cardiologist.

NOTES

- The NT-proBNP level does not differentiate between heart failure due to left ventricular systolic dysfunction and heart failure with preserved ejection fraction
- Obesity, African or African–Caribbean family origin, diuretics, ACE inhibitors, beta-blockers, ARBs and aldosterone antagonists can reduce NT-proBNP levels
- High NT-pro BNP levels can have causes other than heart failure (left ventricular hypertrophy, ischaemia, tachycardia, right ventricular overload, hypoxaemia [including pulmonary embolism], EGFR < 60 ml/minute, sepsis, COPD, diabetes, and cirrhosis)
- NT-proBNP levels increase with age
- NT-proBNP should not be used in primary care to assess or monitor patients with a known diagnosis of chronic heart failure

Management of Chronic Heart Failure

For the management of chronic heart failure, including treatment and monitoring (with information on laboratory tests required as part of management), please see the Derbyshire Joint Area Prescribing Committee (JAPC) local guideline “Management of heart failure with reduced ejection fraction (HFREF)”.

Are there times when specialist secondary care is recommended?

Consider referring for specialist advice in the following situations:

- Refer patients to the specialist multidisciplinary heart failure team in the following situations.
 - Initial diagnosis of heart failure.
 - Management of severe heart failure (NYHA class IV), heart failure that does not respond to treatment, heart failure due to valve disease, or heart failure that can no longer be managed at home.
 - Advice and care of women with known heart failure who are planning a pregnancy or are pregnant. Care of pregnant women should be shared between the cardiologist and obstetrician.

Contacts

Duty Biochemist	01332 789383 (8am to 7pm) 01246 512212 (9am to 5pm) 01283 513313 (8.30 am - 4pm)
On Call Consultant Biochemist	Via UHDB and CRH switchboard
Cardiology Advice	Via UHDB and CRH switchboard

Local guidelines:

Derbyshire Joint Area Prescribing Committee (JAPC) guideline "Management of heart failure with reduced ejection fraction (HFREF)" (Last reviewed April 2022).

National guidelines:

NICE Guideline No 106 (NG106). Chronic Heart Failure in adults: diagnosis and management. September 2018. <http://www.nice.org.uk/NG106>

NICE Clinical Guideline No. 187 (CG187). Acute Heart Failure: diagnosing and managing acute heart failure in adults. November 2021. <http://www.nice.org.uk/CG187>

Patient information website

<http://www.patient.co.uk/health/Heart-Failure.htm>

Document controls

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Approval	Date	Name(s)	Designation
Secondary Care UHDB	08/08/25	Dr N Ahmed	Consultant Cardiologist
Secondary Care CRH	10/08/25	Dr J Cooke	Consultant Cardiologist
Primary Care	10/08/25	Dr P Blackwell	GP
Laboratory (Derbyshire Pathology)	08/08/25	Ms S Knowles	Consultant Clinical Scientist